Neural Networks Primer

Original educational benchmark document for testing document retrieval and grounded question answering.

1. Neurons and Layers

A neural network is a model made of connected computational units commonly called neurons. Each connection has a weight, and a neuron combines its input values with those weights and a bias.

A typical feed-forward neural network contains an input layer, one or more hidden layers, and an output layer. The input layer receives feature values. Hidden layers transform representations. The output layer produces the model's final prediction.

An activation function introduces non-linearity into a network. Without non-linear activation functions, stacking linear transformations would still produce an overall linear transformation.

The ReLU activation function returns zero for negative inputs and returns the input itself for positive inputs. Other activation functions, such as sigmoid and tanh, have different output ranges and behavior.

2. Training a Neural Network

Training adjusts weights and biases to reduce a loss function. The loss function measures how far a model's predictions are from the desired targets for the training examples.

Backpropagation computes gradients of the loss with respect to the network's parameters. An optimizer uses those gradients to update the parameters. Gradient descent is a basic optimization method that moves parameters in a direction intended to reduce the loss.

An epoch is one complete pass through the training dataset. A batch is a subset of training examples processed together before an update. The learning rate controls the size of parameter updates.

If the learning rate is too high, training may become unstable or fail to converge. If it is too low, training may progress very slowly. Learning-rate selection is therefore an important part of model training.

3. Regularization and Evaluation

Regularization refers to methods that discourage a model from fitting the training data too closely. Dropout is a technique that randomly disables a subset of units during training, which can reduce reliance on particular pathways.

Early stopping monitors performance on validation data and stops training when that performance no longer improves. It can help limit overfitting and avoid unnecessary training.

For classification, accuracy is the proportion of predictions that are correct. Precision measures the proportion of predicted positive cases that are truly positive. Recall measures the proportion of actual positive cases that the model identifies.

A confusion matrix summarizes counts of correct and incorrect classification outcomes. Metrics should be selected according to the practical cost of different errors; accuracy alone can be misleading when classes are imbalanced.